Blast from the Past - The Derivative of $\ln(x)$

Let $y = \ln(x)$.

1. Solve this equation for x.

2. Use implicit differentiation to find $\frac{dy}{dx}$ in terms of y.

3. Use questions 1 and 2 to show that $\frac{d}{dx}\ln(x) = \frac{1}{x}$.

Some Stuff from Last Time...

Last time, we looked at inverse trig functions. Today, we will differentiate them...

Questions

1. What are the domain and range of $\arctan x$?

2. If $y = \arctan x$, then _____ $= x$.

Differentiating...

3. Use implicit differentiation on the equation in question 2 to find $\frac{dy}{dx}$ in terms of y.

4. Starting from the identity $\sin^2 y + \cos^2 y = 1$, show that $\tan^2 y + 1 = \sec^2 y$.

5. Show that $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$.

Question Use a similar process to the above, this time using the identity $\sin^2 y + \cos^2 y = 1$ directly to find the derivatives of $\sin^{-1} x$ and $\cos^{-1} x$.

Record your results here:

$$\frac{d}{dx}\sin^{-1} x = _____ \qquad \frac{d}{dx}\cos^{-1} x = _____$$

1. Find the derivatives of the functions below:

   - $g(y) = \sin(\arcsin y)$

   - $p(t) = t \sin^{-1} t$

2. At what value(s) of x does the graph of $\sin^{-1} x$ have vertical tangents?
(Note: show this using the derivative of $\sin^{-1} x$, but recall that we also stated the answer last time when we drew the graph of $\sin^{-1} x$. Can you recall how we got there?)

3. Use the derivative of $\tan^{-1} x$ to show that $\tan^{-1} x$ has horizontal asymptotes as $x \to \infty$ and as $x \to -\infty$. (Hint: If a function has a horizontal asymptote, what must be true of its derivative as $x \to \infty$ or $-\infty$?)

4. Use a linear approximation to estimate arctan(1.1). (Hint: What is arctan(1)?) Is your approximation an overestimate of the true value of arctan(1.1) or an underestimate of it?